# Marco Zappoli

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Master's graduate in Aeronautical Engineering from Politecnico di Milano with a research focus on turbulence, particle-laden flows and computational fluid dynamics. Experienced in high-fidelity numerical simulations and large-scale data analysis, with research resulting in a manuscript submitted to the *Journal of Fluid Mechanics*. Experienced in structured problem-solving and collaborative work through academic projects, with leadership and discipline developed through competitive tennis at Italian second division level.

## Education

**Master of Science in Aeronautical Engineering - Politecnico di Milano (Milan, Italy)** **Sep 2023 - Mar 2026**
- Final mark: 110/110 cum laude
- GPA: 29.47/30
- Thesis: Clustering and preferential sampling in homogeneous shear turbulence
  (Advisor: Dr. Alessandro Chiarini, Co-advisor: Prof. Maurizio Quadrio)
- Relevant courses: Aerodynamics, Turbulence: Physics and Modeling, Numerical Modeling of Differential Problems, Computational Fluid Dynamics, Compressible Fluid Dynamics, Structural Dynamics and Aeroelasticity

**Bachelor of Science in Aerospace Engineering - Politecnico di Milano (Milan, Italy)** **Sep 2020 - Jul 2023**
- Final mark: 105/110
- GPA: 26.77/30

## Publications

**Zappoli, M.**, Quadrio, M. & Chiarini, A. *Clustering and preferential sampling of particles in homogeneous shear turbulence*. Submitted to the *Journal of Fluid Mechanics*, 2026.

## Ongoing Research

**Vortex Stretching and the Turbulent Energy Cascade** **Oct 2026–Present**
Investigating the role of vortex stretching in energy transfer across scales, small-scale dynamics, and intermittency in homogeneous isotropic turbulence.

## Relevant Projects

• **FOSSER: Formation-flight Optimization for a SubSonic mission with Extended Range** **A.Y. 2024/2025**
Analyzed fighter aircraft formation aerodynamics using AVL, optimizing wing design and positioning to reduce drag and extend range, with validation against experimental data and consideration of aeroelastic trade-offs.

• **Experimental Wind Tunnel Testing: Advanced Measurement and Flow Characterization** **A.Y. 2024/2025**
Performed wind tunnel testing on airfoils and eVTOL propellers, including force measurements, pressure and wake analysis, and PIV-based flow field reconstruction using MATLAB.

• **Impact of Ice Formation on the Aerodynamic Performance of the NACA 23012 Airfoil** **A.Y. 2024/2025**
Performed RANS CFD simulations to assess ice accretion effects on airfoil aerodynamics, including flow separation and stall. Developed automated meshing workflows and conducted turbulence/transition model sensitivity analyses.

## Skills

**CAD & CFD Tools:** SolidWorks, Solid Edge, SU2, gmsh
**Programming:** MATLAB, CPL, Python, Fortran
**Tools:** LaTeX, Microsoft Office
**Operating Systems:** Windows, Linux (Ubuntu), High-Performance Computing (CINECA cluster usage)
**Languages:** English (fluent), Italian (native)

## Additional Information

**Certification:** IELTS Academic – Overall 8.0 (L: 8.5, R: 9.0, W: 7.0, S: 7.0); Test date: July 2026
**Membership:** AVIS blood donor - Verbania (2022-present)
**Interests:** Tennis, Formula 1, Fitness, Reading, Cinema, Aviation and Aerospace

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